

APOLLO 13	
LM DATA CARD BOOK	
PART NO.	S / N
SKB32100082 - 387	

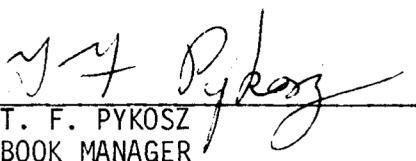
APOLLO 13

FINAL

LM DATA CARD BOOK

MARCH 30, 1970 CHANGE

PREPARED BY:


T. F. PYKOSZ
BOOK MANAGER
TRW TASK 81

MARCH 30, 1970

APPROVED BY: SIGNATURE AUTHORIZED BY TELECOM; MARCH 30, 1970
J. W. O'NEILL
CHIEF
FLIGHT PLANNING BRANCH

It is requested that any organization having comments, questions, or suggestions concerning this document contact T. F. Pykosz, Building 4, Room 278A, Telephone Number HU-3-4015.

This document is under the configuration control of the Crew Procedures Control Board (CPCB). All proposed changes should be submitted to the Apollo Flight Data File Manager, Mr. T. W. Holloway, CF34, Room 230, telephone HU3-4271.

Distribution of this document is controlled by Mr. J. W. O'Neill, Chief, Flight Planning Branch, Flight Crew Support Division.

APOLLO FLIGHT DATA FILE

LM DATA CARD BOOK

BASIC DATE 3/17/70 CHANGE 3/30/70

LIST OF EFFECTIVE PAGES

*Indicates Current Change

PAGE NUMBER	ISSUE
1	.3/24/70
2 thru 7.	.3/17/70
8	.3/24/70
*9 thru 11	.3/30/70
12 thru 14.	.3/17/70

This LOEP reflects changes incorporated
as a result of approved changes:

027, ΔR VALUES

DATE MARCH 24, 1970

LM DATA CARD BOOK

LM ACTIVATION CARD

DAP PAD												
+				+	3	3	7	3	1	LM WT		
+				+	3	7	5	0	0	CSM WT		
+				+	0	0	4	7	6	GMBL		
+				+	0	0	5	7	2			
GYRO TORQUING												
										R1		
										R2		
										R3		
V06 N20												
GET 97 : _____				LM	R1	R2	R3					
				CSM	R1	R2	R3					
GET 98 : _____				LM	R1	R2	R3					
				CSM	R1	R2	R3					
GET _____				LM	R1	R2	R3					
				CSM	R1	R2	R3					

S-BD													
P	(+112)				AOS	(97)				(21)			
Y	(+34)												
P	(+130)				AOS	(99)				(10)			
Y	(+40)												
P	()				AOS	()				()			
Y	()												
AGS													
K FACTOR													
(90) (00) (00) (00)													
+						+	6	0	4	2	7	224	
+						+	2	9	4	0	2	225	
+						+	6	0	4	6	9	226	
-						-	0	1	7	1	8	305	
-						-	5	4	5	0	0	662	
-						-	3	1	7	0	1	673	
												540 (-00002)	
												541 (+00001)	
												542 (-00002)	
												544 (-00006)	
												545 (+00030)	
												546 (+00047)	
UNDOCK/SEP GET													
(99) (16) (21)													

ACTIVATION
GYRO DRIFT COMP

GYRO DRIFT COMPENSATION

ΔT COMPUTATION		
N93 PRO TIME	••	
PREVIOUS TORQ TIME	••	
ΔT	•	
HRS		
BIAS SHIFT		
N93		
BIAS SHIFT		
X	X DEG/HR	
Y	Y DEG/HR	
Z	Z DEG/HR	
GYRO DRIFT		
X	Y	Z
0.0150	0.0030	0.0060
NBD (OLD)		
BIAS SHIFT		
NBD (NEW)		
OCTAL		
PROCEDURES		
V25N01E 1460E NBDX XXXXXE NBDY XXXXXE NBDZ XXXXXE	V21N01E 1460E NBDX XXXXXE E 1461E NBDY XXXXXE E 1462E NBDZ XXXXXE	

DEC/HR	OCTAL		DEC/HR	OCTAL	
	+	-		+	-
000	00000	00000	.765	14532	63245
.015	00177	77600	.780	14731	63046
.030	00377	77400	.795	15131	62646
.045	00576	77201	.810	15330	62447
.060	00775	77002	.825	15527	62250
.075	01174	76603	.840	15726	62051
.090	01374	76403	.855	16126	61651
.105	01573	76204	.870	16325	61452
.120	01772	76005	.885	16524	61253
.135	02171	75606	.900	16723	61054
.150	02371	75406	.915	17123	60654
.165	02570	75207	.930	17322	60455
.180	02767	75010	.945	17521	60256
.195	03166	74611	.960	17721	60056
.210	03366	74411	.975	20120	57657
.225	03565	74212	.990	20317	57460
.240	03764	74013	1.005	20516	57261
.255	04163	73614	1.020	20716	57061
.270	04363	73414	1.035	21115	56662
.285	04562	73215	1.050	21314	56463
.300	04761	73016	1.065	21513	56264
.315	05160	72617	1.080	21713	56064
.330	05360	72417	1.095	22112	55665
.345	05557	72220	1.110	22311	55466
.360	05756	72021	1.125	22510	55267
.375	06155	71622	1.140	22710	55067
.390	06355	71422	1.115	23107	54670
.405	06554	71223	1.170	23306	54471
.420	06753	71024	1.185	23505	54272
.435	07152	70625	1.200	23705	54072
.450	07352	70425	1.215	24104	53673
.465	07551	70226	1.230	24303	53474
.480	07750	70027	1.245	24502	53275
.495	10150	67627	1.260	24702	53075
.510	10347	67430	1.275	25101	52676
.525	10546	67231	1.290	25300	52477
.540	10745	67032	1.305	25477	52300
.555	11145	66632	1.320	25677	52100
.570	11344	66433	1.335	26076	51701
.585	11543	66234	1.350	26275	51502
.600	11742	66035	1.365	26474	51303
.615	12142	65635	1.380	26674	51103
.630	12341	65436	1.395	27073	50704
.645	12540	65237	1.410	27272	50505
.660	12737	65040	1.425	27472	50305
.675	13137	64640	1.440	27671	50106
.690	13336	64441	1.455	30070	47707
.705	13535	64242	1.470	30267	47510
.720	13734	64043	1.485	30467	47310
.735	14134	63643	1.500	30666	47111
.750	14333	63444			

DATE MARCH 17, 1970

LM DATA CARD BOOK

ALTITUDE CHECK/PDI RULES CARD

PDI RULES

- 1. NO ULLAGE-NO GO FOR PDI.
- 2. NO IGNITION DELAY 2 SEC THEN START PB-PUSH; THEN SET DES ENG OVRD-ON AT 5 SEC.
- 3. T/W >1.6 AND DSKY CHANGES >18 fps/2 SEC.
- 4. ATT/RATE <5°/SEC.
- 5. ΔH WITHIN LIMITS >10 SEC AND NOT OUT OF LIMITS >60 SEC.
- 6. DATA GOOD AT >10,000 FT.
- 7. IF NO THROTTLE DOWN BY P64 +15 SEC-ABORT.
- 8. BINGO FUEL 1 MIN 34 SEC AFTER LOW LEVEL OR WHEN FUEL QTY <2% UNLESS LANDING IMMINENT.

NOTE: FOR FLASHING LR ALT OR VEL LIGHTS THAT ARE PRECEDED BY A STEADY LR LIGHT, CYCLE THE RADAR TEST SWITCH.

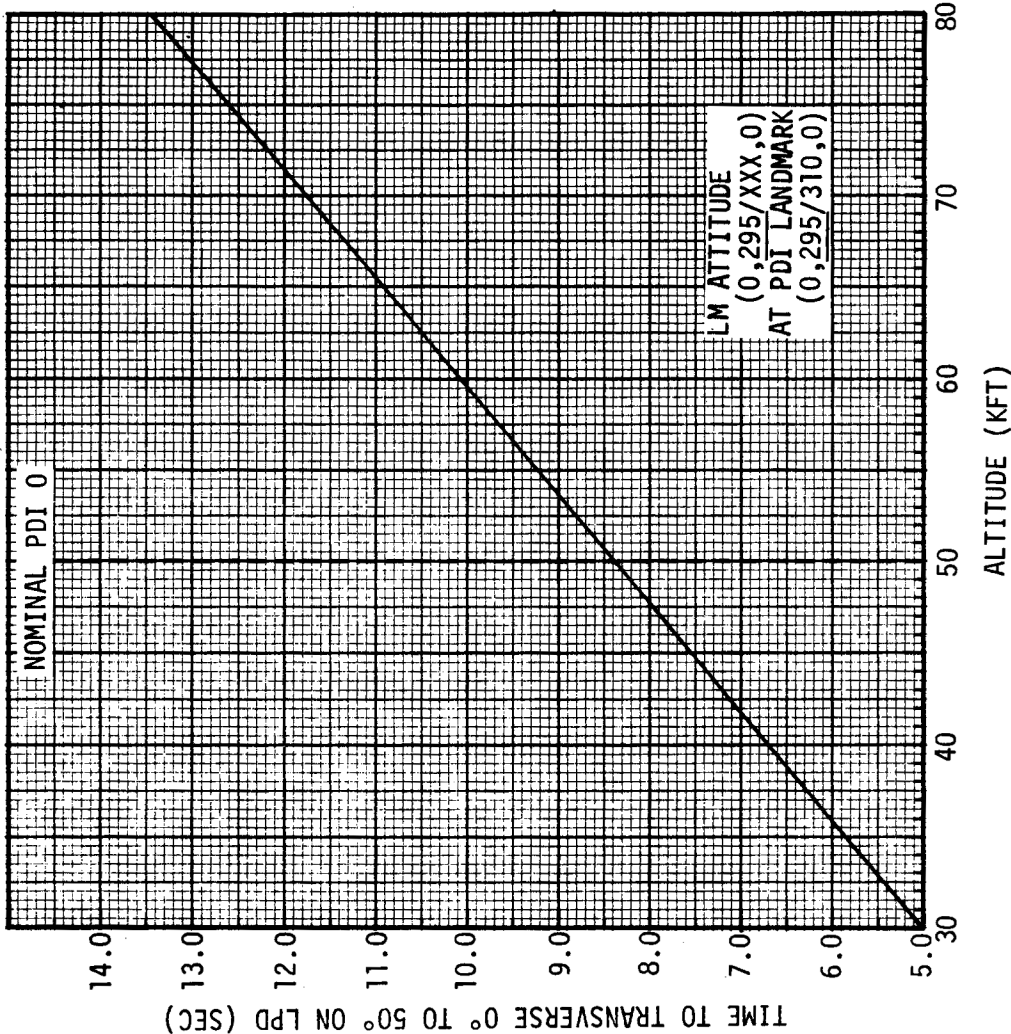
CSM CIRC BURN

P76											
+	0	0		+	0	0	1	0	0	HRS	
										MIN	SEC
										3	5
+	0	0	0	+	0	0	0	3	5	TIG	
+	0			+	0	0	5	0	0	N33	
				+	0	0	6	9	0	N84	
				+	0	0	0	0	0	ΔVX	
				+	0	0	1	4	0	ΔVY	
				+	0	0	1	4	0	ΔVZ	

CSM HA/HP

CMC		LGC		V82	
/		/		OPT 2	

ALTITUDE DETERMINATION CHART



PREPARED BY FPRB/OPS
MISSION APOLLO 13, FEBRUARY 3, 1970

PDI 0/NO PDI + 12 CARD

[illegible]

BURN TIME IF > 1 SEC _____: _____
DATE MARCH 17, 1970

[illegible]

BURN TIME IF > 1 SEC : _____
LM DATA CARD BOOK

APOLLO 13

FLIGHT DATA FILE

T2-1 (PDI 1+20:45) ABORT PAD													
LOG INSERTION GET=													
+ 5 0 0 0													
BOOST GET=													
+ 1 0 0 0 0													
HAM GET=													
+ 5 0 0 0 0													
CSI GET=													
+ 5 0 0 0 0													
L													
HRS N33													
MIN TIG													
SEC													
M													
HRS N37													
MIN TPI													
SEC													
T2-1 AT PDI + 20:45													
N69													
ΔDN RNG													
ΔX RNG													
ΔRLS													
THROTTLE DOWN													
N43													
LAT (+N)													
LONG (+E)													
ALT													

PDI 1/PDI 1 ABORT
PDI 2/PDI 2 ABORT

PDI 1/PDI 1 ABORT
PDI 2/PDI 2 ABORT

PDI 2 ABORT CARD

PDI 2 PAD											
+	0	0			+	0	0	1	0	5	HRS N33
+	0	0	0		+	0	0	0	2	4	MIN PDI
+	0				+	0	1	2	7	0	SEC
X	X	X			X	X					TG0 N61
											CROSSRANGE
X	X	X			X	X	X				R FDAI
X	X	X			X	X	X				P AT TIG
X	X	X			X	X	X				Y
											DEDA 231 IF RQD
(1 <PDI2 ≤8:30) ABORT PAD EARLY											
LOG INSERTION GET= _____ : _____ : _____											
+ _____ 1 0 0 0 0											
BOOST GET= _____ : _____ : _____											
+ _____ 1 0 0 0 0											
HAM GET= _____ : _____ : _____											
+ _____ 1 0 0 0 0											
CSI GET= _____ : _____ : _____											
J											
T1-2 (8:30 <PDI 2 ≤ 15) ABORT PAD LATE											
LOG INSERTION GET= _____ : _____ : _____											
+ _____ 5 5 0 0											
CSI TIG= _____ : _____ : _____											
K											
LOG INSERTION GET= _____ : _____ : _____											
+ _____ 8 8 0 0											
CSI TIG= _____ : _____ : _____											
K											

DATE, MARCH 17, 1970

LM DATA CARD BOOK

DATE MARCH 17, 1970

LM DATA CARD BOOK

APOLLO 13

FLIGHT DATA FILE

PAGE 7

LAUNCH PREP

G&N LUNAR SURFACE CARD

FIRST REV ACTIVITY

N20 _____ OG _____		047 _____ 053 _____		P57, A/T 3 LANDING SITE		377 _____	
_____ IG _____		544 _____ +5:02 _____		N04 _____, _____ TILT		544 _____ +5:02 _____	
_____ MG _____		545 _____		STAR _____		545 _____	
		546 _____				546 _____	
232 _____		465 _____		CURS _____ SPIR _____		514 _____	
P57, A/T 1, REFSMMAT		P57, A/T 2, REFSMMAT		CURS _____ SPIR _____		515 _____	
N04 _____, _____ TILT		STAR1 _____ SPIR _____		V32 _____		516 _____	
N05 _____		CURS _____		CURS _____		047 _____	
N93 _____ X _____ Y _____ Z _____		CURS _____		N05 _____ ANGLE DIFF		053 _____	
P57 A/T 2, REFSMMAT		STAR2 _____		N93 _____ X _____ Y _____ Z _____		NOTES:	
STAR1 _____		CURS _____		P22 ACQ TIME _____ : _____			
CURS _____		CURS _____		P57, A/T 3, LANDING SITE			
V32 _____		CURS _____		N04 _____, _____ TILT			
CURS _____		CURS _____		STAR1 _____			
V32 _____		CURS _____		CURS _____ SPIR _____			
CURS _____		CURS _____		V32 _____			
V32 _____		CURS _____		CURS _____			
CURS _____		CURS _____		N05 _____ ANGLE DIFF			
STAR2 _____		CURS _____		N93 _____ X _____ Y _____ Z _____			
CURS _____		CURS _____		N89 _____ LAT _____			
CURS _____		CURS _____		_____ LONG/2 _____			
V32 _____		CURS _____		_____ ALT _____			
CURS _____		CURS _____		047 _____ 053 _____			
V32 _____		CURS _____		P22 ACQ TIME _____ : _____			
CURS _____		CURS _____					
N05 _____ ANGLE DIFF		CURS _____					
N93 _____ X _____		CURS _____					
_____ Y _____		CURS _____					
_____ Z _____		CURS _____					
N89 _____ LAT _____		CURS _____					
_____ LONG/2 _____		CURS _____					
_____ ALT _____		CURS _____					

LUNAR SURFACE
ABORT/ASCENT

ABORT/ASCENT CARD

DATE. MARCH 24, 1970

LM DATA CARD BOOK

LM ASCENT PAD														
+	0	0					+	0	0	1	3	7	HRS	N33
+	0	0	0				+	0	0	0	0	9	MIN	TIG
+	0						+	0	1	5	8	1	SEC	
+							+	5	5	3	3	8	V (HOR)	
+							+	0	0	3	6	3	V (VERT)	N76
+							+	0	0	0	0	0	*CROSSRANGE	
							+	0	0	0	0	0	047	
							+	4	0	0	0	0	053	
							+	5	8	5	8	1	224/226	
							+	5	6	9	7	8	231	
							+	0	0	3	6	5	465	
+	0	0					+	0	0	1	3	9	HRS	N37
+	0	0	0				+	0	0	0	4	5	MIN	TPI
+	0						+	0	4	0	6	0	SEC	
+							+	1	0	6	8	9	LM WT	
+							+	3	5	6	7	1	CSM WT	
*NOTE: LOAD 8 NM CROSSRANGE IF GREATER THAN 8 NM														
COMMENTS:														
+							HA	V82	+				HA	315
+							HP		+				HP	403
RESIDUALS														
PGNS AGS														
							N85				500			
											ΔVX			
											ΔVY			
											ΔVZ			

BURN RULES

BURN RULES

CRITERIA IS $\dot{X}=3\text{fps}$
PRIORITY OF SOLUTIONS-PGNS, AGS, CMC, CHARTS.

A. RR AGREES WITH VHF WHERE
$$\Delta R = \frac{R}{100} + 0.5\text{NM}, \Delta R \text{ IS ALWAYS } \geq 1\text{NM}$$

IF TWO OF THREE SOLUTIONS AGREE,
BURN THE PRIORITY SOLUTION.

B. RR DOES NOT AGREE WITH VHF.
MSFN ISOLATES FAILED SYSTEM.

C. V90 5fps-NO BURN.

CSI
CDH/PLANE CHANGE

TPI CARD

+	0	0				+	0	0	1	3	9	HRS	N37
+	0	0	0			+	0	0	0	4	5	MIN	TPI
+	0					+	0	4	0	6	0	SEC	
R1(+00000), R2(+02660), R3(+13000) N55													
+	0					+	0	0	2	1	9	ΔVX	N81
+	0					+	0	0	0	0	1	ΔVY	LV
-	0					-	0	0	1	1	0	ΔVZ	
+	0					+	0	3	7	7	7	R	N54
-	0					-	0	1	1	1	5	$\dot{R}$	TIG-5
+	0					+	0	0	2	4	5	$\Delta VF^+/A^-$	N59
+	0					+	0	0	0	0	1	$\Delta VR^+/L^-$	L0S
-	0					-	0	0	0	0	1	$\Delta VD^+/U^-$	
X	X					X	X	0	0	2	2	BT	
307+043.00													

BURN RULES

CRITERIA IS $\dot{X}=2\text{fps}$, $\dot{Y}=5\text{fps}$, $\dot{Z}=6\text{fps}$
 PRIORITY OF SOLUTIONS-PGNS, AGS, CMC, CHARTS.

A. RR AGREES WITH VHF WHERE
 $\Delta R = R + 0.5\text{NM}$, ΔR IS ALWAYS $\geq 1\text{NM}$
 IF TWO OF THREE SOLUTIONS AGREE.
 BURN THE PRIORITY SOLUTION.

B. RR DOES NOT AGREE WITH VHF.
 MSFN ISOLATES FAILED SYSTEM.

RESIDUALS

MAX N49 (0.8,5.0)

PGNS			PGNS			AGS		
PGNS	AGS		ΔVX	N85		ΔVX	500	
			ΔVY			ΔVY	501	
			ΔVZ			ΔVZ	502	
			HA					
			HP					

PGNCS			PGNCS			AGS		
N37 TPI TIG (139:45:41) : : : : :	N58 TPI		N81 TPI		N59 TPI			
	HP (43.3)	$\frac{\Delta V}{(24.7)}$	$\frac{\Delta V}{(21.9)}$	$\frac{\Delta V}{(+0.1)}$	$\frac{\Delta VF^+/A^-}{(24.5)}$	$\frac{\Delta VR^+/L^-}{(+0.1)}$		
373 TIG (+0585.8) 	$\frac{267}{\Delta V} \frac{371}{\Delta V} \frac{TPI+TPF}{TPI}$		CSM SOLUTION (CHANGE SIGN) $\frac{\Delta V}{\Delta V}$		CSM SOLUTION $\frac{\Delta V}{- \Delta V}$			

P76/P27 PADS

P76 PAD													
PURPOSE													
HRS N33													
MIN TIG													
SEC													
ΔVX N84													
ΔVY													
ΔVZ													
PURPOSE													
HRS N33													
MIN TIG													
SEC													
ΔVX N84													
ΔVY													
ΔVZ													
PURPOSE													
HRS N33													
MIN TIG													
SEC													
ΔVX N84													
ΔVY													
ΔVZ													

DATE MARCH 17, 1970

P27 PAD												
V												
V												
V												
INDEX												
01 1173												
02												
03												
04												
05												
06												
07												
10												
11												
12												
13												
14												
15												
16												
17												
20												
21 1213												
22												
23												
24 1216												
X X X												HRS
X X X												MIN
X X X												SEC
0												LAT N43
+ 0												LONG
+ 0												ALT

LM DATA CARD BOOK

AGS SV/IMPACT CARD

[illegible]

AGS SV
IMPACT PAD

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DATE MARCH 17, 1970

LM DATA CARD BOOK

NAJX—MSC

